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ABSTRACT

PersonaVault is a secure digital identity and document management system designed to provide users with a safe and centralized platform for storing and protecting sensitive personal data. In recent years, the rapid growth of digital services and online transactions has increased the risk of data breaches, identity theft, and unauthorized information exposure, highlighting the need for advanced security solutions. Existing systems mainly focus on basic storage or authentication mechanisms, lacking an integrated approach that combines secure data management, intelligent analysis, and proactive threat detection. To address this gap, the proposed system introduces a unified platform that integrates secure authentication, encrypted storage, and intelligent monitoring techniques to enhance data protection. The system implements passwordless authentication using Google OAuth integrated with Supabase, while user data is managed through a structured PostgreSQL-based backend for efficient handling of identities, roles, and metadata. PersonaVault incorporates a secure document vault where users can upload and manage files within protected environments. The system performs document analysis by extracting content from uploaded files and applying Natural Language Processing (NLP) and regular expressions to identify sensitive information such as personally identifiable data and financial details, along with generating unique data fingerprints for tracking. Additionally, the platform integrates automated web scanning to detect unauthorized data exposure by monitoring public sources and cross-referencing stored fingerprints with online content. A risk scoring mechanism is used to evaluate the severity of detected threats, enabling prioritized alerts and better decision-making. By combining strong security practices, intelligent monitoring, and user-friendly design, PersonaVault aims to provide a proactive and scalable solution for digital data protection, enabling users to maintain control over their sensitive information and enhance overall privacy and security.